Extra Dessert #18: Receiving Antennas (RDF, Beverages, Loops)

Beverage antennas, terminating resistors, small loops for direction finding, electrostatic shields, sense antennas, RDF.

Group E9H · 11 pool questions · Self-study reference · Practice exams at eham.net/exams

Beverage Antennas

A Beverage is a long, low, horizontal-wire receiving antenna for the low HF bands. Terminating the far end into a resistor makes it directional toward the termination.

Question	Answer	Q
Design factor for Beverage performance	**It should be at least one wavelength long** (E9H01). Longer is better; ground-coupled traveling-wave behavior.	E9H01
Function of the Beverage's termination resistor	**Absorb signals from the reverse direction** (E9H07). Kills reflections that would otherwise create a bidirectional pattern.	E9H07
Indicator for the correct termination resistance value	**Minimum variation in SWR over the desired frequency range** (E9H06). Flat SWR across the band means the line is properly terminated.	E9H06

160 / 80 m Receiving Strategy

For 160- and 80-meter receiving antennas: Atmospheric noise is so high that directivity is much more important than losses (E9H02). Why a lossy Beverage works fine on low bands: external noise dominates internal loss.

Receiving Directivity Factor (RDF)

RDF = peak antenna gain compared to average gain over the hemisphere around and above the antenna (E9H03). A figure of merit for receive antennas: how much do you favor the wanted direction over noise arriving from the rest of the sky?

Small Loops For Direction Finding

Question	Answer	Q
Challenge of a small wire-loop DF antenna	**It has a bidirectional null pattern** (E9H05). Two nulls 180° apart – you cannot tell which direction the signal came from.	E9H05
Why add an electrostatic shield around a small DF loop	**It eliminates unbalanced capacitive coupling to the antenna's surroundings, improving the depth of its nulls** (E9H04). Shield carries only magnetic-component pickup; nulls become sharp.	E9H04
Function of a sense antenna	**Modifies the pattern of a DF antenna to provide a null in only one direction** (E9H08). Combine a loop with a vertical-whip "sense" antenna and the bidirectional pattern collapses to cardioid.	E9H08
Why a cardioid pattern is useful for DF	**A single null** (E9H11). Cardioid has one deep null behind, so the sharpest null unambiguously points to one bearing.	E9H11
Pattern of a single-turn terminated loop (pennant antenna)	**Cardioid** (E9H09). Pennant antennas are popular small-footprint receive antennas on low HF.	E9H09
Increasing output voltage from a multi-turn receiving loop	**Increase the number of turns and/or the area enclosed by the loop** (E9H10).	E9H10

Dessert 18 Question Bank (E9H)

E9H01. When constructing a Beverage antenna, which of the following factors should be included in the design to achieve good performance at the desired frequency?
A. Its overall length must not exceed 1/4 wavelength.
B. It must be mounted more than 1 wavelength above ground.
C. It should be configured as a four-sided loop.
D. It should be at least one wavelength long.

E9H02. Which is generally true for 160- and 80-meter receiving antennas?
A. Atmospheric noise is so high that directivity is much more important than losses.
B. They must be erected at least 1/2 wavelength above the ground to attain good directivity.
C. Low loss coax transmission line is essential for good performance.
D. All these choices are correct.

E9H03. What is receiving directivity factor (RDF)?
A. Forward gain compared to the gain in the reverse direction.
B. Relative directivity compared to isotropic.
C. Relative directivity compared to a dipole.
D. Peak antenna gain compared to average gain over the hemisphere around and above the antenna.

E9H04. What is the purpose of placing an electrostatic shield around a small-loop direction-finding antenna?
A. It adds capacitive loading, increasing the bandwidth of the antenna.
B. It eliminates unbalanced capacitive coupling to the antenna's surroundings, improving the depth of its nulls.
C. It eliminates tracking errors caused by strong out-of-band signals.
D. It increases signal strength by providing a better match to the feed line.

E9H05. What challenge is presented by a small wire-loop antenna for direction finding?
A. It has a bidirectional null pattern.
B. It does not have a clearly defined null.
C. It is practical for use only on VHF and higher bands.
D. All these choices are correct.

E9H06. What indicates the correct value of terminating resistance for a Beverage antenna?
A. Maximum feed point DC resistance at the center of the desired frequency range.
B. Minimum low-angle front-to-back ratio at the design frequency.
C. Maximum DC current in the terminating resistor.
D. Minimum variation in SWR over the desired frequency range.

E9H07. What is the function of a Beverage antenna's termination resistor?
A. Increase the front-to-side ratio.
B. Absorb signals from the reverse direction.
C. Decrease SWR bandwidth.
D. Eliminate harmonic reception.

E9H08. What is the function of a sense antenna?
A. It modifies the pattern of a DF antenna to provide a null in only one direction.
B. It increases the sensitivity of a DF antenna array.
C. It allows DF antennas to receive signals at different vertical angles.
D. It provides diversity reception that cancels multipath signals.

E9H09. What type of radiation pattern is created by a single-turn, terminated loop such as a pennant antenna?
A. Cardioid.
B. Bidirectional.
C. Omnidirectional.
D. Hyperbolic.

E9H10. How can the output voltage of a multiple-turn receiving loop antenna be increased?
A. By reducing the permeability of the loop shield.
B. By utilizing high impedance wire for the coupling loop.
C. By increasing the number of turns and/or the area enclosed by the loop.
D. All these choices are correct.

E9H11. What feature of a cardioid pattern antenna makes it useful for direction-finding antennas?
A. A very sharp peak.
B. A single null.
C. Broadband response.
D. High radiation angle.

Answer Key

E9H01: D	E9H02: A	E9H03: D	E9H04: B
E9H05: A	E9H06: D	E9H07: B	E9H08: A
E9H09: A	E9H10: C	E9H11: B	